

证明题 已知向量组 $\alpha_1, \alpha_2, \alpha_3$ 线性无关, $\beta_1 = \alpha_1 + \alpha_2, \beta_2 = \alpha_2 + \alpha_3, \beta_3 = \alpha_3 + \alpha_1$,

证明: 向量组 $\beta_1, \beta_2, \beta_3$ 线性无关.

证一: 设有实数 k_1, k_2, k_3 使 $k_1\beta_1 + k_2\beta_2 + k_3\beta_3 = 0$, 即

$$k_1(\alpha_1 + \alpha_2) + k_2(\alpha_2 + \alpha_3) + k_3(\alpha_3 + \alpha_1) = 0,$$

亦即 $(k_1 + k_3)\alpha_1 + (k_1 + k_2)\alpha_2 + (k_2 + k_3)\alpha_3 = 0$.

$\because \alpha_1, \alpha_2, \alpha_3$ 线性无关, $\therefore$ 由线性无关的定义,

$$\begin{cases} k_1 + k_3 = 0, \\ k_1 + k_2 = 0, \\ k_2 + k_3 = 0. \end{cases}$$

因该方程组系数矩阵的行列式 $\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = 2 \neq 0$, 故得唯一解: $k_1 = k_2 = k_3 = 0$,

于是向量组 $\beta_1, \beta_2, \beta_3$ 线性无关. ■

证二: 由于 $\beta_1 = \alpha_1 + \alpha_2, \beta_2 = \alpha_2 + \alpha_3, \beta_3 = \alpha_3 + \alpha_1$, 则

$$(\beta_1, \beta_2, \beta_3) = (\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_3 + \alpha_1) = (\alpha_1, \alpha_2, \alpha_3) \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}. \quad (*)$$

因 $\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = 2 \neq 0$, 所以 $\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$ 可逆. 而可逆矩阵可以化为一系列初

等矩阵的乘积, 于是 (*) 式即是:

矩阵 $(\alpha_1, \alpha_2, \alpha_3)$ 通过一系列的列初等变换化为矩阵 $(\beta_1, \beta_2, \beta_3)$.

由于初等变换不改变矩阵的秩, 所以 $r(\alpha_1, \alpha_2, \alpha_3) = r(\beta_1, \beta_2, \beta_3)$.

$\because \alpha_1, \alpha_2, \alpha_3$ 线性无关, 由线性相关性的判定法则, $r(\alpha_1, \alpha_2, \alpha_3) = 3$. 于

是 $r(\beta_1, \beta_2, \beta_3) = 3$. 故向量组 $\beta_1, \beta_2, \beta_3$ 线性无关. ■

证三：由于 $\beta_1 = \alpha_1 + \alpha_2, \beta_2 = \alpha_2 + \alpha_3, \beta_3 = \alpha_3 + \alpha_1$ ，则

$$(\beta_1, \beta_2, \beta_3) = (\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_3 + \alpha_1)$$

对矩阵 $(\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_3 + \alpha_1)$ 作列初等变换，有

$$\begin{aligned} (\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_3 + \alpha_1) &\xrightarrow{j_1 - j_2} (\alpha_1 - \alpha_3, \alpha_2 + \alpha_3, \alpha_3 + \alpha_1) \\ &\xrightarrow{j_1 + j_3} (2\alpha_1, \alpha_2 + \alpha_3, \alpha_3 + \alpha_1) \\ &\xrightarrow{j_1 \div 2} (\alpha_1, \alpha_2 + \alpha_3, \alpha_3 + \alpha_1) \\ &\xrightarrow{j_3 - j_1} (\alpha_1, \alpha_2 + \alpha_3, \alpha_3) \\ &\xrightarrow{j_2 - j_3} (\alpha_1, \alpha_2, \alpha_3). \end{aligned}$$

上述过程表明，矩阵 $(\beta_1, \beta_2, \beta_3)$ 通过一系列的列初等变换化为矩阵

$$(\alpha_1, \alpha_2, \alpha_3).$$

由于初等变换不改变矩阵的秩，所以 $r(\alpha_1, \alpha_2, \alpha_3) = r(\beta_1, \beta_2, \beta_3)$.

因 $\alpha_1, \alpha_2, \alpha_3$ 线性无关，由线性相关性的判定法则， $r(\alpha_1, \alpha_2, \alpha_3) = 3$. 于是

$r(\beta_1, \beta_2, \beta_3) = 3$. 故向量组 $\beta_1, \beta_2, \beta_3$ 线性无关. ■